

Real-Time Stock Market Prediction Using Sequential Deep Learning Models

Ahsan Iftikhar (22P-9280)

Artificial Neural Networks (AI-3003)

FAST National University of Computer and Emerging Sciences

Faisalabad Campus, Pakistan

Supervisor: Sir Asif Ameer

Email: ahsan.iftikhar.7953@gmail.com

Contributors:

Sir Asif Ameer, Omer Farooq Khan, Aun Ali, Ahsan Butt

Abstract—This project presents a Deep Learning-based stock market prediction system using Sequential Neural Networks including RNN, LSTM, and GRU architectures. Historical stock market data was collected using the Yahoo Finance API. Data preprocessing techniques including normalization and sequence generation were applied before training the models.

Three sequential deep learning architectures were implemented and evaluated using RMSE metrics. Experimental results demonstrated that the GRU model achieved the best performance with the lowest prediction error. A Streamlit-based frontend application was also developed for real-time stock prediction.

Index Terms—Deep Learning, RNN, LSTM, GRU, Stock Market Prediction, Sequential Models, Streamlit, TensorFlow

I. INTRODUCTION

Stock market prediction is one of the most challenging problems in financial analysis because stock prices are highly dynamic and affected by multiple factors. Deep Learning models have recently shown promising results in time-series forecasting and sequential data analysis.

This project focuses on predicting stock market closing prices using Sequential Deep Learning models including Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Historical stock market data was collected using Yahoo Finance API.

The main objective of this project is to compare the performance of different sequential neural network architectures and identify the most effective model for stock price prediction.

II. DATASET OVERVIEW

The dataset was collected using the Yahoo Finance API. Apple (AAPL) stock market data was used in this project.

The dataset contains:

- Open Price
- High Price
- Low Price
- Close Price
- Volume

The dataset covers stock market records from 2018 to 2025. Data preprocessing steps included:

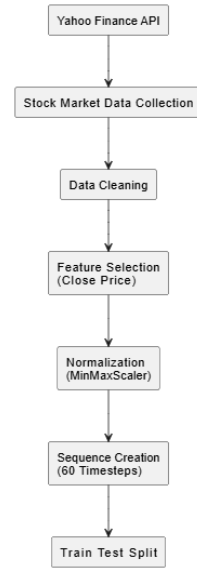


Fig. 1. Stock Market Data Pipeline

- Data cleaning
- Feature selection
- MinMax normalization
- Sequence generation using 60 timesteps
- Train-test splitting

III. METHODOLOGY

Three sequential deep learning architectures were implemented and evaluated.

A. RNN Model

The Recurrent Neural Network (RNN) model was implemented for sequential stock prediction. RNN models are capable of processing sequential information but suffer from vanishing gradient problems during long sequence learning.

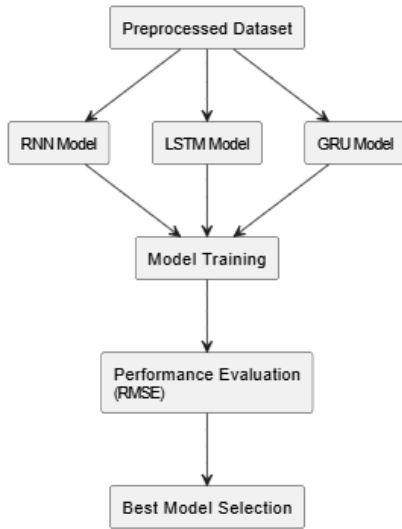


Fig. 2. Model Training Pipeline

B. LSTM Model

Long Short-Term Memory (LSTM) networks improve sequential learning by using memory cells and gating mechanisms. LSTM models are effective for capturing long-term dependencies in stock market data.

C. GRU Model

Gated Recurrent Unit (GRU) is a lightweight sequential architecture similar to LSTM but with fewer parameters and faster computation. GRU achieved the best prediction performance in this project.

IV. SYSTEM ARCHITECTURE

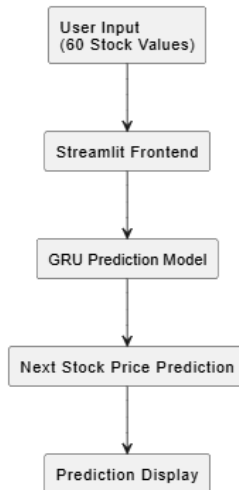


Fig. 3. Inference System Architecture

A Streamlit-based frontend application was developed for stock prediction. Users can input normalized stock values, and the trained GRU model predicts the next stock price.

Stock Market Prediction App

This application predicts the next stock price using a trained GRU Deep Learning model.

Enter Last 60 Normalized Stock Prices

Enter 60 comma-separated values

0.33,0.35,0.36,0.38,0.40,0.42,0.43,0.44,0.46,0.47,
0.49,0.50,0.52,0.53,0.55,0.56,0.58,0.60,0.61,0.63,
0.64,0.65,0.66,0.68,0.69,0.70,0.71,0.72,0.73,0.74,
0.75,0.76,0.77,0.78,0.79,0.80,0.81,0.82,0.83,0.84,
0.85,0.86,0.87,0.88,0.89,0.90,0.91,0.92,0.93,0.94

Predict Next Price

Predicted Normalized Stock Price: 0.8934

Fig. 4. Streamlit Frontend Interface

V. EXPERIMENTS AND RESULTS

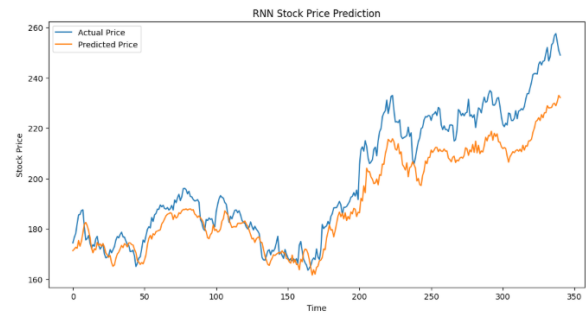


Fig. 5. RNN Prediction Results

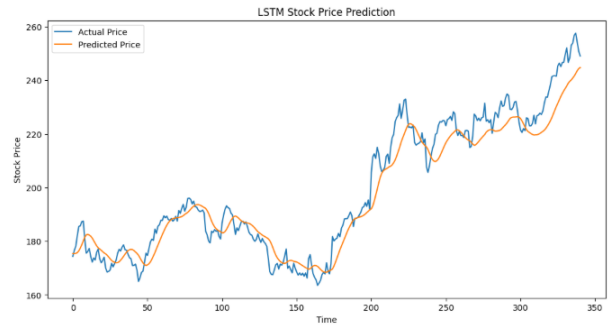


Fig. 6. LSTM Prediction Results

The models were evaluated using Root Mean Square Error (RMSE).

TABLE I
MODEL COMPARISON

Model	RMSE
RNN	11.17
LSTM	6.59
GRU	5.36

Experimental results demonstrated that the GRU model achieved the best performance with the lowest prediction error.

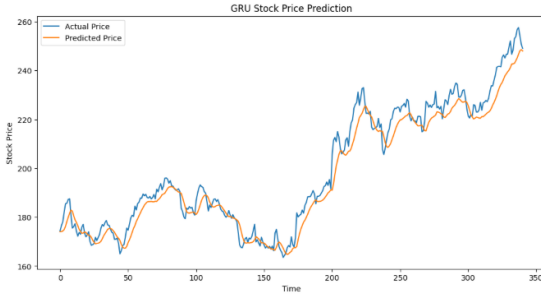


Fig. 7. GRU Prediction Results

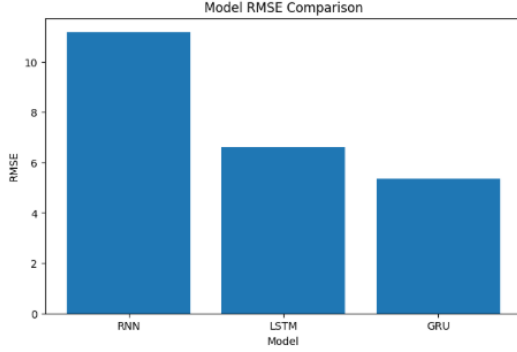


Fig. 8. RMSE Comparison Graph

VI. CHALLENGES FACED

Several implementation challenges were encountered during the development of this project.

- Sequential stock prediction required careful preprocessing and sequence generation.
- Selecting the appropriate timestep length affected model performance.
- RNN models suffered from higher prediction error due to limited long-term memory.
- Training deep learning models required significant computational resources.
- Hyperparameter tuning was necessary to improve model generalization.

VII. CONCLUSION

This project presented a Sequential Deep Learning-based stock market prediction system using RNN, LSTM, and GRU architectures.

Experimental results showed that the GRU model achieved the best prediction performance with the lowest RMSE value. The developed Streamlit application successfully demonstrated real-time stock prediction using Deep Learning techniques.

Future improvements may include multi-feature forecasting, real-time stock streaming, and transformer-based architectures for improved prediction accuracy.

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